CareQuest

Product Requirements Document

Long-term care health monitoring with gamified engagement

Field	Detail
Version	0.1 (Draft)
Owner	Product
Status	For review
Metrics in scope	Blood pressure, blood glucose, cholesterol (lipid panel)

1. Overview

CareQuest is a web app for long-term care facilities and home-care programs. It tracks patients' blood pressure (BP), blood glucose (BG) and cholesterol (lipid panel), alerts clinicians when readings go out of range, and uses games and rewards to keep patients engaged in their care routines.

Problem

- Chronic-condition monitoring in long-term care is often manual, paper-based, and inconsistent.
- Patients disengage from routine measurements, causing data gaps and late detection of deterioration.
- Clinicians lack a single view of trends across multiple metrics and patients.

Vision

A single, accessible platform where care teams see risk early and patients want to take part in their own monitoring.

Goals

- Increase measurement adherence to >=85% of scheduled readings.
- Reduce time to clinician notification for critical readings to under 5 minutes.
- Achieve >=60% weekly active use of games among eligible patients.
- Give clinicians a trend view per patient in under 3 clicks.

Non-goals (v1)

- Diagnosis or automated treatment decisions
- Medication ordering or prescribing
- Real-money rewards or gambling mechanics
- Native mobile apps (responsive web only)

2. Users and Personas

Persona	Needs
Patient (often 65+, may have low vision, tremor, or mild cognitive impairment)	Simple, large-type interface; easy logging; fun, low-pressure games
Nurse / Caregiver	Fast data entry, ward-level dashboard, prioritized alerts
Physician / Care Manager	Trends, targets, reports, care-plan adjustments
Family member (optional, with consent)	Read-only summary and encouragement messages
Facility Admin	User management, audit logs, compliance reports, reward budget

3. Scope of Health Metrics

Metric	Unit	Typical capture	Frequency
Blood pressure (systolic/diastolic, pulse)	mmHg, bpm	Bluetooth cuff or manual	1-3x/day per care plan

Metric	Unit	Typical capture	Frequency
Blood glucose	mg/dL or mmol/L	Bluetooth glucometer, CGM import, or manual; tagged fasting / pre-meal / post-meal / bedtime	1-4x/day per care plan
Cholesterol (Total, LDL, HDL, Triglycerides)	mg/dL	Lab result entry or import (HL7/FHIR); point-of-care device	Every 3-12 months

Note: Cholesterol is measured infrequently, so its UX centers on lab-result timelines and "next test due" reminders rather than daily logging.

4. Functional Requirements

Priority: P0 = MVP, P1 = fast follow, P2 = later.

4.1 Accounts and Access

ID	Pri	Requirement
FR-1	P0	Role-based access control (patient, caregiver, clinician, family, admin).
FR-2	P0	Secure login with MFA for staff; simplified login for patients (PIN, QR badge, or caregiver-assisted).
FR-3	P0	Patient consent management, including per-family-member sharing permissions.
FR-4	P0	Full audit log of data views and edits.

4.2 Data Capture

ID	Pri	Requirement
FR-5	P0	Manual entry with input validation (plausible ranges, unit checks, duplicate detection).
FR-6	P1	Bluetooth/API integration with common BP cuffs and glucometers.
FR-7	P1	Lab result import via FHIR/HL7 or PDF upload with clinician verification.
FR-8	P0	Every reading stores timestamp, source (device/manual), entered-by, and context tags (fasting, after meal, resting, etc.).
FR-9	P1	CGM data import (e.g., summary metrics like time-in-range).

4.3 Monitoring and Alerts

ID	Pri	Requirement
FR-10	P0	Per-patient configurable target ranges and alert thresholds, set by a clinician (guideline-based defaults, editable).
FR-11	P0	Three alert tiers: Informational, Warning, Critical. Critical alerts (e.g., severe hypoglycemia, hypertensive crisis range) notify assigned staff immediately via in-app, email and SMS/push.
FR-12	P0	Alert acknowledgment and escalation: unacknowledged critical alerts escalate to the next contact after a configurable time.
FR-13	P0	Missed-reading reminders and overdue-lab notifications.
FR-14	P1	Trend-based alerts (e.g., 7-day upward BP trend).

4.4 Dashboards and Reporting

ID	Pri	Requirement
FR-15	P0	Ward/facility dashboard: patients sorted by risk, open alerts, overdue measurements.
FR-16	P0	Patient detail view: time-series charts per metric with target bands and annotations for medication or care-plan changes.
FR-17	P0	Patient-friendly summary: large-type "traffic light" status and plain-language messages.
FR-18	P1	Exportable reports (PDF/CSV) for clinical reviews and family meetings.
FR-19	P2	Population analytics for facility quality improvement.

4.5 Gamification and Rewards

Design principle: Rewards are earned for participation and effort (measuring on time, completing routines, engaging with education), never for achieving "good" readings. Rewarding results would encourage falsifying or hiding numbers and could disadvantage patients whose conditions are harder to control.

ID	Pri	Requirement
FR-20	P0	Points system: points for logging a reading on schedule, completing a daily check-in, and finishing health-education mini-lessons.
FR-21	P0	Streaks and gentle progress indicators with a "grace day" so one missed day doesn't erase progress. No punitive messaging.
FR-22	P0	Game library (3-4 games in v1) designed for older adults: memory match; word search / scramble; simple puzzles and health-knowledge trivia; a daily "garden" or "journey" builder where check-ins grow a virtual scene.
FR-23	P0	Games unlock by completing the day's monitoring tasks, as a "thank you"; always optional.
FR-24	P0	Facility-configurable reward catalog: badges, virtual items, themes, and facility-provided real rewards (choice of activity, extra visit time, diet-appropriate treats, charity donation in the patient's name).
FR-25	P1	Care-team and family encouragement (cheers, stickers); opt-in team challenges rather than head-to-head rankings.
FR-26	P1	Difficulty adapts to patient ability; per-patient game and reward settings for clinicians and caregivers.
FR-27	P0	Per-patient opt-out from gamification without losing monitoring features.
FR-28	P0	Clinical values never appear on any leaderboard or public view. Leaderboards, if enabled, show only participation points and are opt-in.
FR-29	P1	Dietary/medical guardrails: physical rewards filtered by each patient's care plan (e.g., no sweets for patients with diabetes).

4.6 Care Plan and Communication

ID	Pri	Requirement
FR-30	P1	Care plan tasks and schedules per patient, driving reminders.
FR-31	P1	Clinician notes and secure messaging within a patient record.
FR-32	P2	Medication log and correlation with readings.

5. Key User Flows

- Daily patient flow: Login, see today's checklist (large buttons), log or sync a reading, get immediate friendly feedback and points, unlock a game.
- Critical reading flow: Reading submitted, threshold check, critical alert to assigned nurse, acknowledgment, note/action recorded. The patient sees a calm message ("Your nurse is coming to check in with you").
- Clinician review flow: Dashboard, select patient, review trends, adjust targets or care plan, save with audit trail.

6. Non-Functional Requirements

Area	Requirement
Accessibility	WCAG 2.2 AA; minimum 18px base font with size controls; high contrast; touch targets >=48px; screen-reader support; voice input option; no reliance on color alone; no time-pressured game mechanics
Performance	Page loads <2s on typical facility Wi-Fi; alert delivery <60s from submission
Availability	99.9% uptime; offline-tolerant data entry that syncs when connection returns
Security	Encryption in transit (TLS 1.2+) and at rest; least-privilege access; session timeouts suited to shared devices; regular penetration testing
Privacy and compliance	HIPAA (US), GDPR (EU) as applicable; BAAs with vendors; data retention and deletion policies; minimum-necessary access
Interoperability	FHIR R4 APIs; ability to connect with EHR systems
Localization	English at launch; unit switching (mg/dL / mmol/L); multi-language ready
Auditability	Immutable logs of clinical data changes and alert handling

7. Regulatory and Clinical Safety

- Regulatory status must be determined early. If the app provides automated interpretation or alerts that drive clinical decisions, it may qualify as Software as a Medical Device (FDA in the US, MDR in the EU). Engage regulatory counsel before build.
- Alert thresholds and defaults must be reviewed and signed off by a clinical advisory board.
- The app supports, and does not replace, clinician judgment. Patient-facing messages must not give diagnoses or medication advice.
- Clear on-screen guidance: "If you feel unwell, tell your nurse or call emergency services."
- Clinical risk assessment (hazard analysis) before launch, with a plan for handling missed or failed alerts.

8. Success Metrics

Category	Metric	Target
Adherence	% of scheduled readings completed	>=85%
Clinical response	Median time to acknowledge critical alerts	<5 min
Engagement	Weekly active game users among opted-in patients	>=60%
Retention	90-day patient retention in the program	>=80%

Category	Metric	Target
Satisfaction	Patient face-scale rating / staff SUS score	>=4/5 / >=75
Workload	Staff time per reading logged	-30% vs. baseline
Safety	Missed critical alerts	0

Clinical outcome measures (e.g., improved time-in-range, BP control rates) are tracked in a pilot study rather than used as launch gates.

9. Risks and Mitigations

Risk	Mitigation
Patients falsify readings to earn rewards	Reward participation only; device-verified readings where possible; anomaly detection
Alert fatigue among staff	Tiered alerts, tunable thresholds, digest for low-priority items
Low digital literacy / cognitive decline	Caregiver-assisted mode, minimal UI, extensive usability testing with target users
Games feel patronizing	Co-design with older adults; adult-appropriate themes; opt-out
Data breach	Encryption, access controls, monitoring, incident response plan
Regulatory reclassification	Early legal review; modular architecture separating monitoring from analytics
Device integration complexity	Start with manual entry plus 1-2 supported devices

10. Release Plan

Phase	Timeline	Scope
MVP (Pilot)	Months 0-4	RBAC, manual entry, target ranges, tiered alerts, dashboards, points/streaks, 2 games, reward catalog
v1.0	Months 5-8	Device integrations, lab import, family portal, 3-4 games, reports, care-plan tasks
v1.5+	Months 9-12	Trend alerts, adaptive difficulty, group challenges, EHR integration, analytics

Pilot: 1-2 facilities, 30-60 patients, 12 weeks, with clinician and patient feedback loops.

11. Open Questions

1. Which regulatory pathway applies, and what is the intended-use statement?
2. Which devices and EHR systems must be supported at launch?
3. Who funds physical rewards, and what budget limits apply per patient?
4. What consent model is needed for family access and for patients with impaired capacity?
5. Should patients with severe cognitive impairment be excluded from gamification, or use a caregiver-mediated version?
6. Is offline mode required at launch for facilities with unreliable connectivity?

12. Suggested Tech Approach (Non-binding)

- Frontend: Responsive web (React/TypeScript), accessibility-first component library
- Backend: REST/GraphQL API, FHIR-compatible data model, relational database
- Notifications: Push/SMS/email provider with delivery receipts
- Infrastructure: HIPAA-eligible cloud, managed secrets, centralized logging